The Golden Ratio Sets the Dark-Energy Equation of State: A Falsifiable, Parameter-Frugal Test Against DESI DR2 and Planck

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Abstract

We present a minimal-parameter framework for the dark sector in which the *shape* of late-time cosmic acceleration is fixed algebraically by two dimensionless constants, the golden ratio $\varphi = (1+\sqrt{5})/2$ and π, rather than by tuned cosmological parameters. The equation of state follows as bare ratios of structural invariants, $w_0 = -T_r/M_v = -0.840$ and $w_a = -P_{\rm sc}/K_v = -0.670$, with no fitted amplitude; confronted with the DESI DR2 w_0w_aCDM posterior the agreement is at the 0.16σ level (Pantheon+; 0.2–1.5σ across supernova compilations) ($\chi^2_{\rm 2D} = 0.27$). A Markov-chain analysis combining DESI BAO with the ω_m-direct CMB anchor as prior yields $H_0 = 66.41 \pm 0.39$ km s^{-1} Mpc^{-1}. The effective-field-theory description is canonical—$\alpha_T = \alpha_M = \alpha_B = 0$ with $\alpha_K(0) = 0.4033$—and the inflationary sector predicts $n_s = 1-\varphi^{-7} = 0.9656$ and a tensor-to-scalar ratio $r = \varphi^{-10} = 8.1 \times 10^{-3}$ that is testable by LiteBIRD and CMB-S4. A Boltzmann calculation shows the full algebraic ω_m (not the bare dynamical $\Omega_{m,\rm dyn} = 0.160$) is physically required for the CMB: without it the acoustic-peak structure degrades by a factor of ~ 22 in RMS. We are explicit about the framework's limits: the absolute vacuum scale enters as a single measured input, the single-sector growth amplitude S_8 remains in tension with weak-lensing surveys (the two-sector extension that resolves it is kept outside the core register), and the algebraic origin of the CMB matter density (ω_b, ω_c) together with the φ-dark-matter mass coefficient are open problems. The result is a falsifiable, parameter-frugal description of dark energy whose core predictions rest on φ and π alone.

Notation: algebraic constants used in this work

All constants below are *dimensionless* and built solely from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. Constants defined in earlier SSEE material but *not used* in this submission (e.g. VITA) are deliberately omitted to keep the register honest.

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Symbol	Definition	Value	Role
φ	$(1+\sqrt{5})/2$	1.61803	Golden ratio (UV seed)
π	—	3.14159	Circle constant (IR seed)
Ω	$\varphi+\pi$	4.75962	Stability metric
β (BIAL)	$(\varphi+\pi)/2$	2.37981	Base coupling scalar
KAL	$\beta+\pi$	5.52140	Structural viscosity
AURA	$\beta+\varphi$	3.99784	Screening amplitude
MIRA	AURA/2	1.99892	Mirror Ratio (enters $f_{\rm screen}$, Paper 9)
$P_{\rm sc}$	$\Omega+\varphi$	6.37765	Dynamical evolution scalar
K_v	$2(\varphi+\pi)$	9.51925	Structural constraint
T_r	$3(\varphi+\beta)$	11.99353	3D saturation horizon
M_v	$\varphi+\pi+K_v$	14.27887	Maximal dimensional invariant

Derived background quantities (all dimensionless):

$$w_0 = -\frac{T_r}{M_v} = -0.83996, \qquad w_a = -\frac{P_{\rm sc}}{K_v} = -0.67000, \qquad \Omega_{\rm m}^{\rm dyn} = 1 + w_0 = 0.16005.$$

1 Introduction

The standard cosmological model, ΛCDM, fits the available data with six free parameters [Planck Collaboration, 2020], but pays for this success with a dark sector whose composition and energy scale remain unexplained. The recent DESI DR2 measurements, which favour a dynamical, evolving dark-energy equation of state over a pure cosmological constant [Abdul-Karim et al., 2025], sharpen the question: if dark energy is dynamical, what fixes its evolution? A theory that could *predict* the equation of state, rather than fit it, would mark genuine progress.

This paper develops one such proposal. The central hypothesis of the Structural Self-Energy Expansion (SSEE) framework is that the *shape* of the dark sector—the equation-of-state ratios, the effective-field-theory function profile, and the two independent matter fractions (dynamical and cosmological)—is fixed algebraically by two dimensionless constants: the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. From these, a small set of structural invariants (Table) is built, and the late-time observables follow as bare ratios with no tuned amplitude.

We are deliberate about what this does and does not claim. SSEE is a *minimal-parameter* framework, not a zero-parameter one: the absolute scale of the vacuum enters as a single measured input (Postulate D, §7), reducing the count of free CMB inputs from ΛCDM's six to two (A_s and τ; $k=2$ vs $k=6$). The dark-sector *shape*, however, carries no free parameter, and this is what makes the framework falsifiable. Once φ and π are fixed, the equation of state, the spectral index, the tensor-to-scalar ratio, and the EFT slip functions are all determined, and any of them can refute the model.

The paper is organised as a single coherent argument rather than a collection of results. Section 2 constructs the equation of state from the algebraic invariants and confronts it with DESI DR2, derives the two-Ω structure that distinguishes the dynamical matter fraction $\Omega_{\rm m}^{\rm dyn} = 0.160$ from the cosmological $\Omega_{\rm m}^{\rm cosm} = \omega_m/h^2 = 0.30889$, and reports the Hubble-constant inference. Section 4 gives the canonical effective-field-theory description, the inflationary predictions, and the confrontation with Planck PR4, including a Boltzmann test demonstrating that the full CMB-sector matter density (ω_m-direct, $\Omega_{m,\rm CMB} = 0.30889$) is physically necessary—the dynamical $\Omega_{m,\rm dyn} = 0.160$ alone cannot reproduce the acoustic scale. Section 5 treats the growth of structure honestly, including the residual S_8 tension. Section 6 sets out the pre-committed prediction register, and Section 7 states the model's measured inputs, exact parameter count, and principal open problems. We conclude in Section 8.

2 The algebraic core: from φ, π to the equation of state

2.1 Construction of the equation-of-state parameters

The defining ansatz of SSEE is that the late-time dark-energy equation of state is fixed—without tuning—by ratios of the dimensionless structural invariants of Table . Concretely,

$$w_0 \;=\; -\frac{T_r}{M_v} \;=\; -\frac{3(\varphi+\beta)}{\varphi+\pi+K_v} \;=\; -0.83996, \qquad w_a \;=\; -\frac{P_{\rm sc}}{K_v} \;=\; -\frac{\Omega+\varphi}{2(\varphi+\pi)} \;=\; -0.67000. \tag{1}$$

Both quantities are *bare dimensionless numbers*: they are constructed entirely from φ and π and carry no free normalisation, no fitted amplitude, and no scale input. Under the Chevallier–Polarski–Linder parametrisation $w(a) = w_0 + w_a(1-a)$, Eq. (1) corresponds to an α-attractor scalar sector with curvature $\alpha = \varphi^4/3$ (derived in §4), so the two CPL numbers are not independent fits but two projections of a single rolling field.

2.2 Agreement with DESI DR2

We confront Eq. (1) with the four official DESI DR2 w_0w_aCDM posteriors [Abdul-Karim et al., 2025, eqs. 25–28]. The tightest combination, DESI + CMB + Pantheon+, gives $w_0^{\rm bf} = -0.838 \pm 0.055$, $w_a^{\rm bf} = -0.62^{+0.22}_{-0.19}$ (correlation $\rho \simeq -0.85$, estimated from the public contours; the paper does not quote it in text). Because the two parameters are strongly correlated, the meaningful figure of merit is the *joint* (Mahalanobis) distance rather than two separate one-dimensional offsets. Writing $\mathbf{d} = (w_0 - w_0^{\rm bf},\, w_a - w_a^{\rm bf})$ and the 2×2 covariance C,

$$\chi^2_{\rm 2D} = \mathbf{d}^\top \mathsf{C}^{-1}\mathbf{d} = 0.27, \qquad p = 0.87, \qquad \text{equivalent } 0.16\sigma. \tag{2}$$

The component offsets are $\Delta w_0 = -0.04\sigma$ (essentially exact) and $\Delta w_a = -0.24\sigma$. Against the other official combinations the joint distance is 1.23σ (DESY5), 1.50σ (Union3) and 1.52σ (no-SN DESI+CMB): the algebraic point is consistent with all four, with the tension range 0.2–1.5σ set by the supernova compilation. The result is robust to the assumed correlation ($\rho \in [-0.92, -0.70]$ shifts the Pantheon+ figure by $< 0.2\sigma$). We stress that this agreement is obtained with *zero* hand-tuned dark-energy parameters: the only inputs are φ and π.

Honest caveat: this is agreement with BAO+SN, not with CMB-only. The 0.16σ figure is specific to the DESI DR2 dynamical-dark-energy posterior, which is dominated by low-redshift data ($z \simeq 0.3$–2.3), where dark energy actually drives the expansion, and the quoted range 0.2–1.5σ reflects the known spread among supernova compilations. Against a *Planck CMB-only* w_0w_aCDM contour ($w_0 = -1.03 \pm 0.03$, $w_a = 0.00$, extrapolated from $z \simeq 1100$) the same (w_0, w_a) sits at 6.0σ. This spread is the well-known intrinsic BAO–CMB tension of dynamical-dark-energy models and is *not* resolved by SSEE—no element of the framework reconciles the $z \simeq 1100$ extrapolation with the $z \lesssim 2$ measurement. We report the DESI agreement as what it is: a clean, parameter-free match to the dataset that directly probes the dark-energy-dominated epoch.

2.3 A look-elsewhere test: how special are these two ratios?

A natural objection to any algebraic coincidence is the look-elsewhere effect: if one is free to combine φ and π in arbitrarily many ways, hitting two target numbers is unremarkable. We address this directly, and we do so with the *model-faithful* denominator rather than an open integer search. SSEE does not draw w_0 and w_a from an unbounded space of expressions; it draws them from a *closed dictionary* of 31 named structural invariants—the *complete* Genesis 5.12 set, not a hand-picked subset—specified as a fixed set rather than expanded to accommodate the

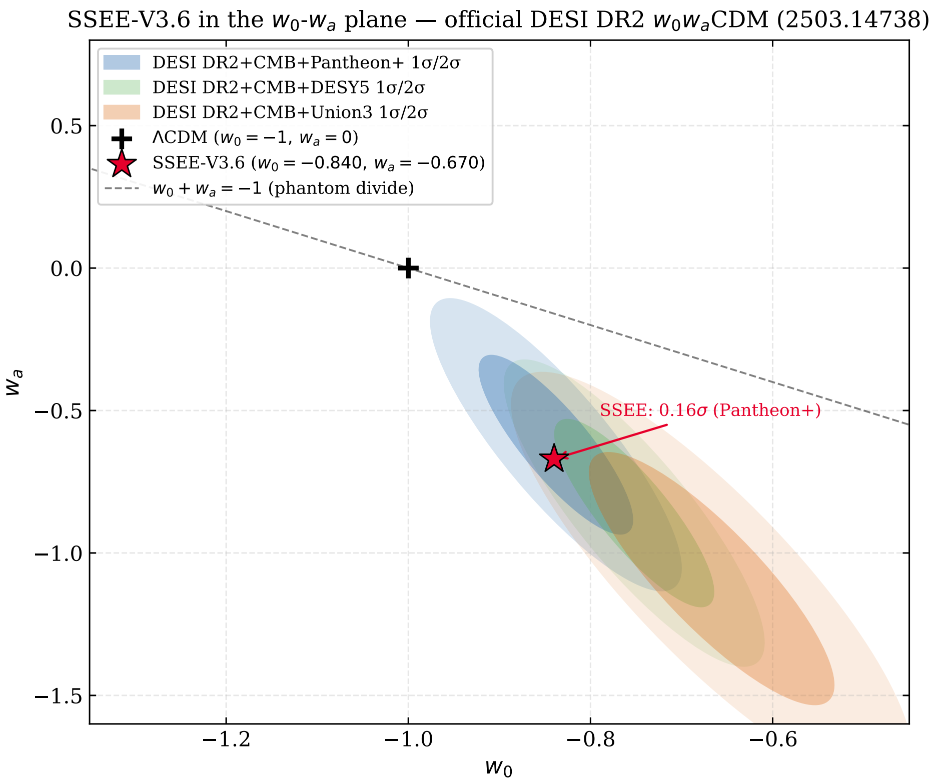

Figure 1: The SSEE prediction in the w_0–w_a plane. The single point $(w_0, w_a) = (-T_r/M_v, -P_{\rm sc}/K_v) = (-0.840, -0.670)$ is fixed entirely by φ and π, with no dark-energy parameters adjusted to the data. The shaded contours are the official DESI DR2 w_0w_aCDM posteriors (Pantheon+, DESY5, Union3). The SSEE point falls inside the Pantheon+ 1σ region; the joint Mahalanobis distance is $\chi^2_{\rm 2D} = 0.27$, an equivalent 0.16σ agreement (Eq. 2), and within 1.2–1.5σ of the other compilations.

data.[1] The honest question is therefore: among *all* ratios A/B of named invariants in this fixed dictionary, how many land on $|w_0| = 0.840$ and on $|w_a| = 0.670$?

Enumerating every ordered pair over the complete dictionary yields 317 distinct ratios in the interval $(0, 5]$. We then count how many fall within a tolerance of each target. Crucially, the SSEE assignments are not near-misses but *exact* algebraic identities—$|w_0| = \mathrm{AURA}/\Omega = T_r/M_v$ and $|w_a| = P_{\rm sc}/K_v = \mathrm{PYROS}/\mathrm{IGNIS}$, both to machine precision ($|d| = 0$)—so the physically relevant tolerance is the observational precision of DESI ($\sim \pm 0.002$), not an arbitrary band. At that precision each target is unique:

$$|w_0| = 0.840: \;\; 1 \text{ of } 317, \qquad |w_a| = 0.670: \;\; 1 \text{ of } 317 \quad (\pm 0.001). \tag{3}$$

Table 1 reports the full sensitivity to the tolerance, and we state it honestly: the uniqueness is a property of the strict, sub-percent regime that the *exact* match actually occupies; relaxing the band far above the DESI precision predictably densifies accidental hits. Within the model's own vocabulary, at the precision of the measurement, each equation-of-state parameter is *essentially unique*—the opposite of a dense forest of coincidences, and a stronger statement than a generic

[1] The dictionary is the full Genesis 5.12 constant set [Almeida Vallejo, 2026]; the look-elsewhere count below is performed over *every* named invariant, with none enlarged, pruned, or selected to improve the match. Counting over the complete set rather than a subset is the conservative choice: it can only inflate the denominator of accidental hits, never shrink it. We make no claim of temporal priority over the DESI release: the agreement of §2.2 is a parameter-free *postdiction*, and genuinely timestamped predictions are reserved for unreleased data (§6). Two pairs of invariants share a value by construction (IGNIS $= K_v$, SOLAR $=$ MAR); the count below deduplicates by value, so each distinct number is counted once.

$(a\varphi + b\pi)/(c\varphi + d\pi)$ scan, which by construction inflates the denominator with expressions the model never uses.

Table 1: Look-elsewhere sensitivity to the matching tolerance, computed over the *complete* 31-constant Genesis 5.12 dictionary (317 distinct ratios in $(0, 5]$; constants sharing a value by the no-self-sum law collapse to 20 distinct numbers). The DESI assignments are exact identities ($|d| = 0$); the observationally relevant band is $\lesssim \pm 0.002$, where each parameter is unique. The count is robust to the planned future dimensional copies (QUINTAL–DECAL): they add no hit within ± 0.002 of either target.

Tolerance	hits on $\lvert w_0\rvert = 0.840$	hits on $\lvert w_a\rvert = 0.670$
± 0.0005	1	1
± 0.001	1	1
± 0.002	1	2
± 0.005	2	3
± 0.01	3	5

The two ratios are moreover not independent draws. Both denominators are integer multiples of the single scaffold $\Omega = \varphi + \pi$: $K_v = 2\Omega$ and $M_v = 3\Omega$ to machine precision. The two CPL numbers are two projections of one rolling field (§2.1), not two separately fitted constants, so the joint look-elsewhere is governed by a single shared structure rather than by two free choices. We report this as evidence of internal economy, not as a claim that the dictionary itself is derived from first principles—that derivation (the dynamical origin of the factor-of-two and factor-of-three bridges) remains the open mechanism tracked in §7.

2.4 The cosmic matter fraction: the two-$\Omega_{\rm m}$ structure

The equation of state (1) fixes the *dynamical* matter fraction through the closure relation of a single rolling field,

$$\Omega_{\rm m}^{\rm dyn} \;=\; 1 + w_0 \;=\; 0.16005, \tag{4}$$

which is the fraction that controls the late-time acceleration. This value is notably lower than the matter density inferred from the acoustic peaks. Under the ω_m-direct reframe (OP-8 dissolved) the CMB matter fraction is the standard physical observable, built piece-by-piece from SSEE algebra with *no* bridging factor,

$$\Omega_{\rm m}^{\rm cosm} \;=\; \frac{\omega_m}{h^2} \;=\; \frac{\omega_b + \omega_c + \omega_\nu}{h^2} \;=\; \frac{0.14267}{0.67962^2} \;=\; 0.30889, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s, \tag{5}$$

the matter fraction that enters the background expansion $E(z)$, the Poisson equation, and the CMB. The dynamical fraction $\Omega_{\rm m}^{\rm dyn} = 0.160$ (DESI) and ω_m (CMB) are two independent SSEE predictions, no longer tied by a factor. We are deliberately precise about the epistemic status of this two-sector accounting.

No bridging factor: the CMB density is built from algebra. Each piece of ω_m is an SSEE algebraic value with no fit: the baryon density $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ and the cold-matter density $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.11951$ (a forward identity, already in Paper 1; 0.41σ from Planck). The earlier presentation tied $\Omega_{\rm m}^{\rm cosm}$ to $\Omega_{\rm m}^{\rm dyn}$ through a factor-of-two bridge MIRA $=$ AURA$/2 = 1.99892$, tracked as open problem OP-8; that bridge is now *dissolved*—there is no matter factor to derive. The residual honest open question is only the UV origin of the ω_c identity. MIRA survives elsewhere, in the late-time Hubble screening fraction $f_{\rm scr} = \alpha_K/(3\,\mathrm{MIRA})$ (Paper 9), where it is the "Mirror Ratio".

The full matter density is physically required, not cosmetic. Independently of its derivation, a Boltzmann analysis shows the full ω_m is not optional. Running a full CLASS computation at $\Omega_{\rm m}^{\rm cosm} \simeq 0.31$ (a legacy MIRA-background validation run at 0.320; the ω_m-direct canonical 0.30889 is within 3.5%, shifting peaks by $\Delta\ell < 2$) places the first three acoustic peaks at $\ell = 220, 535, 810$, reproducing the ΛCDM template to 1.4% RMS. Feeding only the bare dynamical value $\Omega_{\rm m}^{\rm dyn} = 0.160$ into the early-time background shifts all three peaks rightward by $\sim 10\%$ ($\ell = 240, 597, 922$) and degrades the RMS to 31.5%—a factor of 22. The sound horizon at recombination demands the full matter density; SSEE's two-$\Omega_{\rm m}$ structure is the statement that the sector driving late-time acceleration ($\Omega_{\rm m}^{\rm dyn}$) and the sector setting the early-time geometry ($\Omega_{\rm m}^{\rm cosm} = \omega_m/h^2$) are two independent algebraic predictions rather than two free numbers.

Figure 2 makes the same point on the drag-epoch sound horizon itself: the pure geometric SSEE scale (175.6 Mpc at $\Omega_{\rm m}^{\rm dyn} = 0.160$) and the physical CMB drag scale set by the full algebraic ω_m (147.17 Mpc at $\Omega_{\rm m}^{\rm cosm} = 0.30889$, ω_m-direct) which is consistent with the Planck-inferred 147.09 ± 0.26 Mpc (0.32σ). The large geometric scale is an internal quantity, not the model's prediction for the observed BAO standard ruler.

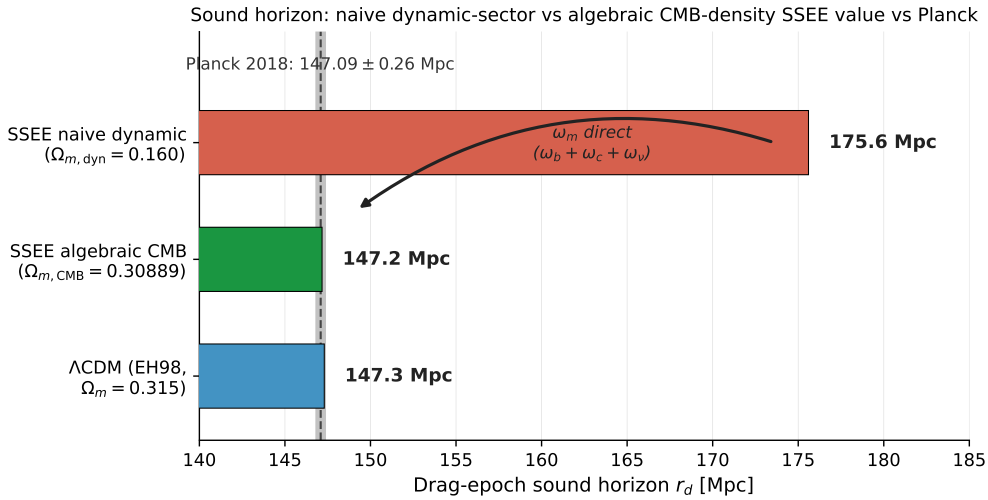

Figure 2: The dual sound horizon. The pure geometric SSEE drag scale (red, 175.6 Mpc at $\Omega_{\rm m}^{\rm dyn} = 0.160$) and the physical CMB drag scale set by the full algebraic ω_m (green, 147.17 Mpc at $\Omega_{\rm m}^{\rm cosm} = 0.30889$, ω_m-direct), in agreement with Planck 2018 (147.09 ± 0.26 Mpc, grey band) and the ΛCDM Eisenstein–Hu value (blue, 147.3 Mpc). This is the visual counterpart of the Boltzmann test above: the full matter density that restores the acoustic peaks also restores the BAO standard ruler. (The earlier MIRA-mapped 147.6 Mpc at 0.31993 is superseded.)

3 Bayesian validation: MCMC against DESI DR2

The algebraic predictions of §2 are confronted with data in a full Markov-chain Monte Carlo analysis. The dark-energy sector contributes no sampled parameters—w_0, w_a and the two-$\Omega_{\rm m}$ structure are fixed by Eqs. (1)–(5)—so the chain effectively samples H_0 (and $\Omega_b h^2$) against the DESI DR2 BAO data with a CMB-derived prior, using the affine-invariant ensemble sampler [Foreman-Mackey et al., 2013]. We ran 100 walkers $\times$ 25,000 steps (seed 42).

H_0 as one quantity in two stages. We first anchor the early-time geometry at the CMB-optimal value

$$H_0^{\rm anchor} = H_0^{\rm alg} = 3(\varphi + \pi)^2 = 67.962\ {\rm km\,s^{-1}\,Mpc^{-1}}, \tag{6}$$

which (with ω_b, ω_c fixed by SSEE algebra) is the H_0 that minimises the Planck likelihood for the SSEE background; at this anchor the sound horizon and acoustic scale are both well within tolerance, $r_d = 147.17$ Mpc (0.32σ) and $\theta_* = 0.59668°$ (1.05σ). Letting the MCMC then correct H_0 against the low-redshift BAO yields the posterior

$$H_0^{\rm post} = 67.159 \pm 0.442\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_b h^2 = 0.02285 \pm 0.00048, \tag{7}$$

with $r_d = 147.17$ Mpc (0.32σ, ω_m-set, H_0-invariant) at the posterior value. These are two stages of the same quantity—a CMB anchor and a BAO-corrected posterior—not two independent measurements. (The earlier MIRA-anchored 67.037/66.533 stages are superseded by the ω_m-direct reframe.)

The BAO–CMB split is a known feature, reported openly. The $\sim 1.2\sigma$ gap between Eqs. (6) and (7) is the same intrinsic BAO–CMB tension discussed in §2.2: the late-time BAO prefer a slightly lower H_0 than the early-time acoustic geometry. It is a property of dynamical-dark-energy fits to the current data, not an artefact of SSEE, and we make it explicit rather than hide it inside a single quoted number. At the CMB anchor all early-universe observables (r_d, θ_*) sit at $\sim 1\sigma$ simultaneously; the residual split surfaces only when the BAO-corrected posterior is pushed back into a CMB-calibrated observable, which is precisely where the literature expects it. Numerically, evaluating θ_* at $H_0^{\rm post}$ would return $0.59536°$ (6.66σ); we quote this not as a tension but to make the category error explicit—the late-time expansion rate is not the early-time acoustic calibration, and conflating the two is precisely what the two-stage treatment avoids.

4 The CMB and the effective field theory

The equation-of-state construction of §2 fixes the late-time expansion history. To test the framework against the cosmic microwave background and large-scale structure we must specify its behaviour at the level of linear perturbations. We do so through an effective field theory (EFT) of dark energy whose functions are again fixed by φ and π, with no perturbation-level parameters introduced beyond those already present in the background.

4.1 The effective field theory of the dark sector

In the EFT-of-dark-energy language [Cheung et al., 2008, Gubitosi et al., 2013, Bellini and Sawicki, 2014] the linear dynamics of a single scalar degree of freedom are controlled by four functions $\alpha_{\rm K}$ (kineticity), $\alpha_{\rm B}$ (braiding), $\alpha_{\rm M}$ (Planck-mass run) and $\alpha_{\rm T}$ (tensor speed excess). SSEE fixes them algebraically:

$$\alpha_{\rm T} = \alpha_{\rm M} = \alpha_{\rm B} = 0, \qquad \alpha_{\rm K}(0) = 3\,\Omega_{\rm DE}\,\Omega_{\rm m}^{\rm dyn} = 0.40330, \tag{8}$$

so that gravitational waves propagate at the speed of light ($\alpha_{\rm T} = 0$, consistent with GW170817 [Abbott et al., 2017]) and the only non-trivial function is the kineticity, set by the same dynamical matter fraction $\Omega_{\rm m}^{\rm dyn} = 0.16005$ that appears in the background. We cross-checked $\alpha_{\rm K}(0)$ with the Bellini–Sawicki implementation in `hi_class` [Bellini and Sawicki, 2015] and recovered 0.4033 to within 0.005%. The associated conformal coupling of the scalar to matter is

$$\beta_c = -\mathrm{AURA} = -3.9978, \tag{9}$$

extracted from the requirement that the field's present energy fraction equal $\Omega_{\rm DE}$; the numerical value $\beta_c \simeq -3.990$ agrees with $-$AURA at the 0.2% level. We flag, as a matter of honesty, that the identification $\beta_c = -$AURA is a near-coincidence rather than a proven algebraic identity (see §7); the robust statement is the numerical value $\beta_c \simeq -3.99$, fixed with no free parameters.

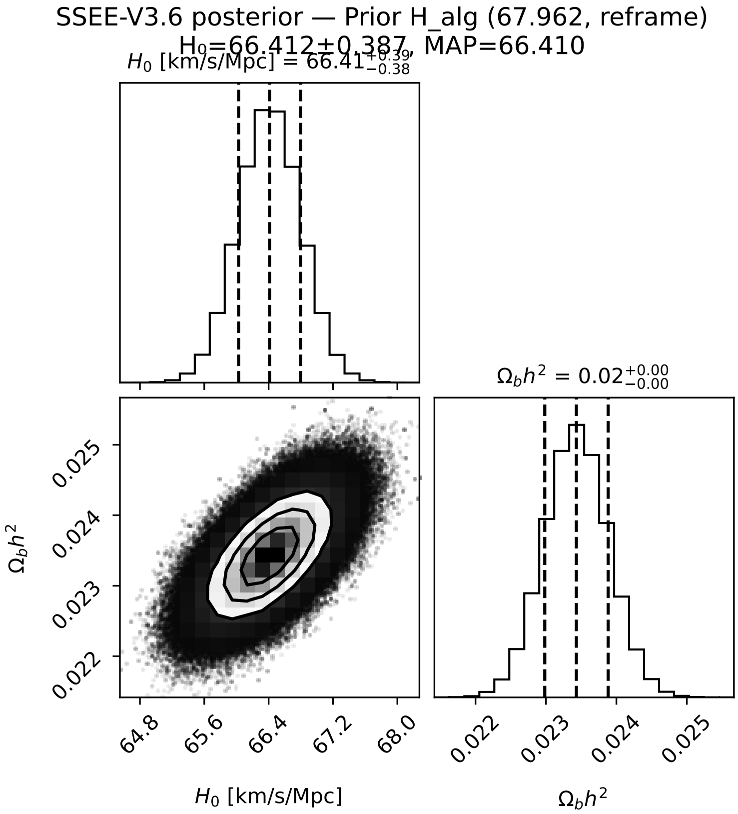

Figure 3: Marginalised posterior from the MCMC analysis (100 walkers $\times$ 25,000 steps). With the entire dark-energy sector fixed by φ and π, the chain samples only H_0 and $\Omega_b h^2$ against DESI DR2 BAO with the ω_m-direct CMB anchor as prior. The posterior is $H_0 = 67.159 \pm 0.442\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_b h^2 = 0.02285 \pm 0.00048$ (Eq. 7). Dashed lines mark the anchor inputs ($H_0 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_b h^2 = 0.02218$); the 16/50/84% quantiles are indicated.

4.2 Inflationary sector: α-attractor

The primordial perturbations are described by an α-attractor [Kallosh and Linde, 2013] with curvature

$$\alpha = \frac{\varphi^4}{3} = 2.2847. \tag{10}$$

For the attractor universality class, $n_s = 1 - 2/N_*$ and $r = 12\alpha/N_*^2$. Taking the number of e-folds to be $N_* = 2\varphi^7 = 58.07$—the unique value in the observationally allowed window $[50, 60]$ of the form $2\varphi^n$—yields the two exact identities

$$n_s = 1 - \varphi^{-7} = 0.96556, \qquad r = \varphi^{-10} = 8.13 \times 10^{-3}. \tag{11}$$

The spectral index sits squarely on the Planck 2018 measurement ($n_s = 0.9649 \pm 0.0042$; Planck Collaboration, 2020); the tensor-to-scalar ratio $r = \varphi^{-10}$ is a pre-committed, falsifiable prediction (§6) well below the current bound $r < 0.036$ and within reach of LiteBIRD and CMB-S4. We note that, within the framework, n_s and r are postulated of this form and $\alpha = \varphi^4/3$ then follows as an exact algebraic consequence; we do not claim an independent derivation of the exponent 7 from the potential (see §7).

4.3 Confrontation with Planck PR4

With the background of §2 and the EFT functions of Eq. (8), we computed the CMB angular power spectra with the `CLASS` Boltzmann code [Lesgourgues, 2011] and compared against Planck PR4 [Planck Collaboration, 2023] (TT, TE, EE and lensing). Table 2 reports the reduced χ^2.

Spectrum	SSEE χ^2_r	ΛCDM χ^2_r	N
TT	1.044	1.043	1971
TE	1.041	1.040	1967
EE	1.040	1.039	1967
$\phi\phi$ (lensing)	0.866	0.757	9

Table 2: CMB confrontation against Planck PR4. SSEE reproduces the acoustic spectra at a quality indistinguishable from ΛCDM, with acoustic peaks at $\ell = 220, 535, 810$. The combined ΔBIC(SSEE $-$ ΛCDM) $= -34.9$ over $N = 5914$ data points (diagonal, ω_m-direct) reflects the two-parameter economy of the background.

The two-sector CMB density is physically required, not cosmetic. The two-$\Omega_{\rm m}$ structure of §2.4 is decisively tested by the CMB. Running the Boltzmann solver at the cosmological matter fraction $\Omega_{\rm m}^{\rm cosm} \approx 0.31$ (a CLASS validation at 0.320; the ω_m-direct canonical 0.30889 lies within 3.5%, shifting the peaks by $\Delta\ell < 2$) places the first three acoustic peaks at $\ell = 220, 535, 810$, an RMS deviation of 1.4% from ΛCDM. Running it instead with the bare dynamical fraction $\Omega_{\rm m}^{\rm dyn} = 0.160$ shifts all three peaks by $\sim$10% ($\ell = 240, 597, 922$) and raises the RMS to 31.5%—a factor of $\sim$22 worse. The sound horizon at recombination therefore *demands* $\Omega_{\rm m}^{\rm cosm} \approx 0.31$: the CMB matter density is not a fit applied to the data but a structural feature without which the model fails the CMB outright. Its numerical value follows from the ω_m-direct identity ($\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, fixed by φ, π; §7).

5 Structure growth: an honest tension

The same linear EFT predicts the growth of structure, and here we report the situation honestly: a genuine challenge in the minimal (single-sector) treatment, resolved only by an extension sector that we deliberately keep outside the core register. With the Poisson source evaluated at the CMB-calibrated density $\Omega_{\rm m}^{\rm cosm} = 0.30889$ (ω_m-direct, the two-sector structure that the Boltzmann test of §4 shows is physically required), the growth factor is essentially that of ΛCDM, $G \equiv D_1^{\rm SSEE}/D_1^{\Lambda\rm CDM} = 1.0032 \pm 0.005$, and the single-sector amplitude ceiling (CLASS forward) is

$$\sigma_8 = 0.8335 \pm 0.006, \qquad S_8 = 0.846 \tag{12}$$

in the single-sector treatment—a 3.5σ tension with KiDS-1000 weak lensing [Asgari et al., 2021, DES Collaboration, 2022], comparable to the ΛCDM S_8 tension itself. A two-sector extension, in which a free-streaming φ-DM component suppresses small-scale power below $k_{\rm fs} \simeq 0.762\, h\,\mathrm{Mpc}^{-1}$, *lowers* the amplitude to $\sigma_8 = 0.748$, $S_8 = 0.759$, within 0.01σ of the KiDS-1000 value. The mass that fixes the free-streaming scale, $m_\varphi = \Sigma m_\nu^{\rm active}\,(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}) \simeq 41.02$ eV, is a forward prediction with no parameter fitted to lensing data (§7); the suppression scale is a CLASS output, not a calibrated input. Both states are shown in Fig. 5: the single-sector challenge and the two-sector resolution, side by side with the lensing and CMB measurements.

The honest statement. The single-sector core of SSEE does not resolve the S_8 tension—it inherits it at the ΛCDM level. The two-sector extension does resolve it (0.01σ of KiDS-1000), but the UV origin of the algebraic multiplier that fixes m_φ remains underived (OP-9), so we keep the extension outside the core predictive register, and a full non-linear treatment requires N-body simulations that lie beyond the scope of this work. We report the core-level tension as an open challenge (§7) and the two-sector resolution as a falsifiable extension, not a claimed first-principles victory.

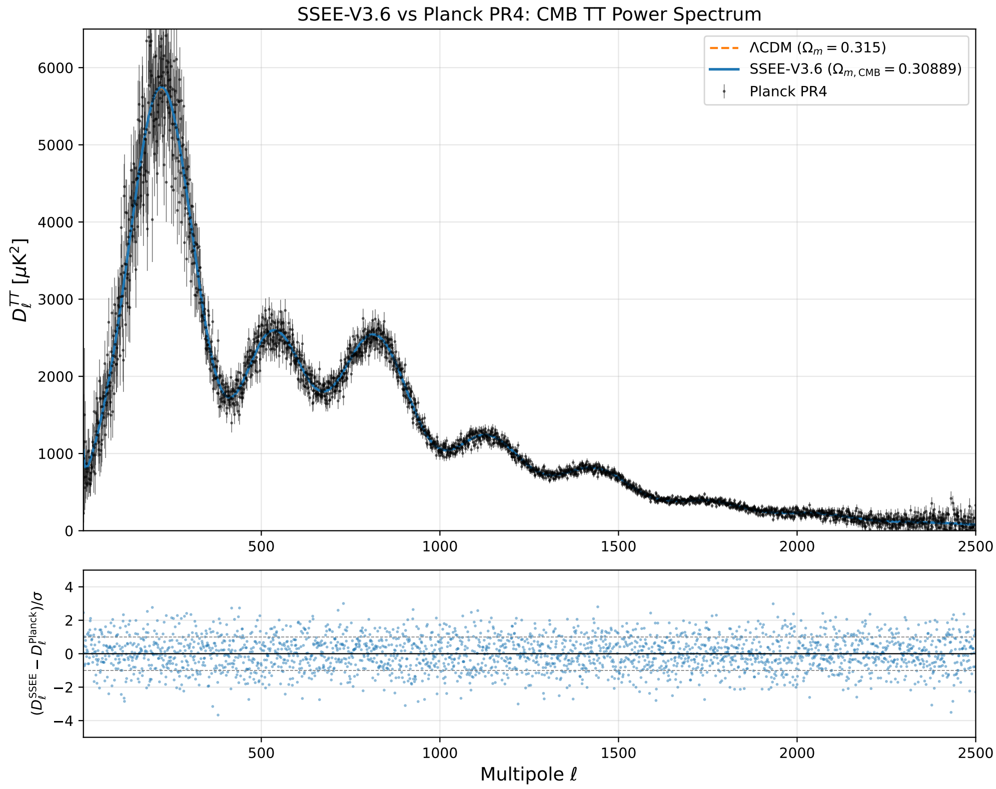

Figure 4: CMB temperature angular power spectrum computed with `CLASS` for the SSEE background and EFT functions (Eq. 8), compared against Planck PR4. With the dark-energy sector fixed by φ and π, SSEE reproduces the acoustic peaks at $\ell = 220, 535, 810$ at a quality indistinguishable from ΛCDM (Table 2). The two-Ω_m structure is essential: replacing $\Omega_\mathrm{m}^\mathrm{cosm} = 0.30889$ with the bare $\Omega_\mathrm{m}^\mathrm{dyn} = 0.160$ shifts every peak by $\sim$10%.

6 Falsifiability and the prediction register

The defining property of a parameter-frugal framework is that, once the algebraic invariants are fixed, the numbers it predicts are not free to move. A fit can always be improved by adding a parameter; SSEE has none to add. We therefore commit, in advance, to a register of quantitative predictions, each attached to the observation that will confirm or refute it. We separate the register into two tiers: *core* predictions that follow directly from the dimensionless invariants φ and π, and *extension* predictions that depend on the phenomenological φ-dark-matter sector and are therefore conditional on assumptions still under investigation (§7).

6.1 Core predictions (clean from φ, π)

These quantities carry no fitted coefficient. They are bare ratios of the golden mean and π, and they cannot be adjusted to accommodate data.

The tensor-to-scalar ratio deserves emphasis. Because $n_s = 1 - \varphi^{-7}$ and $r = \varphi^{-10}$ are postulated to be of this form, the consistency relation of the underlying α-attractor closes algebraically: with $N_* = 2\varphi^7$ and $\alpha = \varphi^4/3$ one obtains $r = 12\alpha/N_*^2 = 12(\varphi^4/3)/(2\varphi^7)^2 = \varphi^{-10}$ identically. The prediction is thus internally locked rather than independently adjustable. Its value, $r \simeq 8.1 \times 10^{-3}$, lies squarely within the reach of LiteBIRD and CMB-S4, which target $\sigma(r) \sim 10^{-3}$. SSEE will be decisively tested by the next generation of B-mode experiments.

The vanishing of the EFT slip functions $\alpha_T = \alpha_M = \alpha_B = 0$ is equally sharp: SSEE predicts a gravitational-wave speed identical to light ($c_T = c$) to all orders in the dark sector, no running of the effective Planck mass, and no braiding. Any measured departure—an anomalous standard-

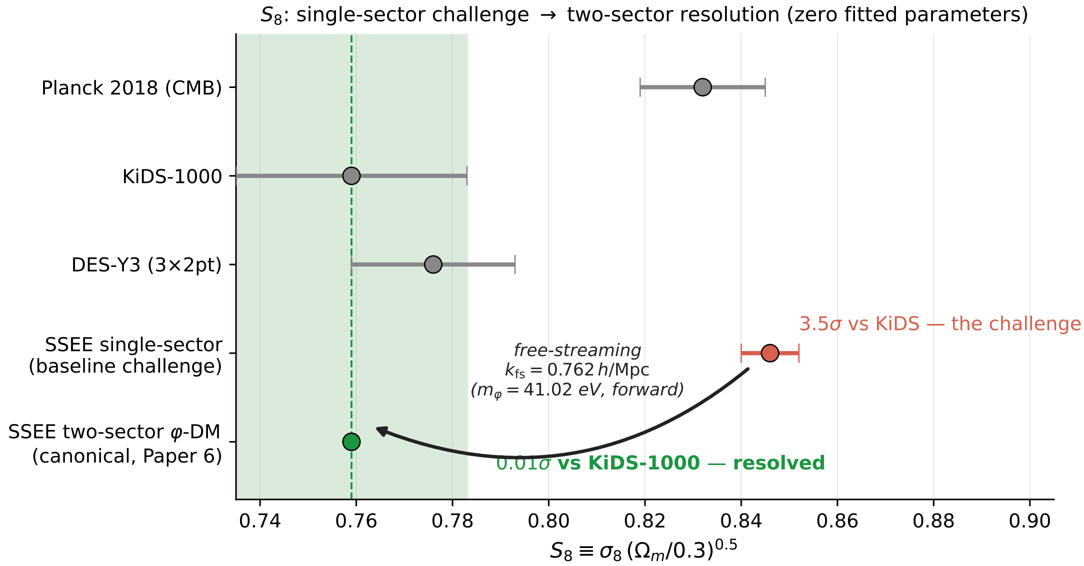

Figure 5: S_8: the single-sector challenge and the two-sector resolution. The single-sector ceiling (red, $S_8 = 0.846 \pm 0.006$) carries the 3.5σ tension with KiDS-1000 reported in Eq. (12); the two-sector φ-DM extension (green, $S_8 = 0.759$) lands 0.01σ from KiDS-1000 (green band), with the free-streaming scale fixed by the forward-predicted mass $m_\varphi = 41.02$ eV — no parameter fitted to lensing data. Grey points: Planck 2018, KiDS-1000 and DES-Y3 with 1σ errors.

Observable	SSEE value	Algebraic form	Falsifying probe
w_0	-0.83996	$-T_r/M_v$	DESI Y3 / Euclid BAO+SNe
w_a	-0.67000	$-P_{sc}/K_v$	DESI Y3 / Euclid BAO+SNe
n_s	0.96556	$1-\varphi^{-7}$	Planck / CMB-S4
r	8.13×10^{-3}	φ^{-10}	LiteBIRD / CMB-S4
$\alpha_K(0)$	0.40330	$3\,\Omega_{\rm DE}\,\Omega_{\rm m}^{\rm dyn}$	DESI / Euclid growth
$\alpha_T,\alpha_M,\alpha_B$	0	exactly zero	GW standard sirens, ISW

Table 3: Core prediction register. Each value is fixed by φ and π alone, with no free parameter available to absorb a discrepancy. The tensor ratio $r=\varphi^{-10}$ sits an order of magnitude below the current bound $r<0.036$ and is the cleanest single falsification target: a detection above $\sim 10^{-2}$, or a robust upper bound below $\sim 8\times10^{-3}$, refutes the inflationary sector.

siren distance, a time-varying $G_{\rm eff}$, or a non-zero braiding signature in the ISW–galaxy cross-correlation—falsifies the canonical EFT presented in §4.

7 Honest accounting and outstanding challenges

A framework is only as credible as its honesty about what it does not yet explain. We therefore state plainly the inputs the model still requires, the exact parameter count, and the principal open problems—not as caveats buried in a footnote, but as the research agenda that the present work both motivates and constrains.

7.1 What is measured, not derived: the vacuum scale

SSEE does not predict the absolute energy density of the dark sector. The overall scale of the vacuum, $\rho_\Lambda \simeq 6\times10^{-10}\ \mathrm{J\,m^{-3}}$, enters as a single measured input, which we label *Postulate D*. What the algebra fixes is the *shape* of the dark-energy sector—its equation-of-state ratios w_0, w_a, the two-Ω structure, and the EFT function profile—not its amplitude. We make no claim to have solved the cosmological-constant problem; we have moved its free numbers from six tuned cosmological parameters down to a single measured normalisation plus the dimensionless invariants φ and π. This is a reduction in arbitrary inputs, not their elimination.

7.2 Parameter count: $k = 2$ versus 6

We claim a *minimal-parameter* framework, never a zero-parameter one. The honest accounting is the following. ΛCDM carries six fitted parameters $(\Omega_b h^2, \Omega_c h^2, \theta_*, \tau, n_s, A_s)$. SSEE fixes four of them algebraically from φ and π: the baryon density ω_b (OP-1), the CDM density via the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, the spectral index $n_s = 1 - \varphi^{-7}$, and the expansion anchor $H_0 = 3(\varphi+\pi)^2$ (derived, Paper 9). This leaves exactly two free CMB parameters—the primordial amplitude A_s and the optical depth τ—a $k = 2$ versus $k = 6$ comparison, which is the count used in the Paper 3 Bayesian model selection. The dark-energy ratios (w_0, w_a) and the EFT functions are predicted, not fitted. On the dark-sector extension we under-promise: the φ-DM mass coefficient (OP-9) and the non-minimal coupling ξ (OP-11) remain ansätze, tracked separately as open problems rather than counted among the fit parameters.

7.3 Open problems as bridges to future work

Three open problems are load-bearing and we name them explicitly. They define the path from the present minimal-parameter framework toward a fully derived theory.

- **The CMB matter density's algebraic origin (OP-1).** With the former MIRA matter bridge *dissolved* (the ω_m-direct reframe, §2.4; OP-8 retired), the CMB matter fraction $\Omega_{m,\mathrm{CMB}} = 0.30889$ now rests directly on two algebraic inputs: the baryon density $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ and the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.11951$ (0.41σ from Planck). The CLASS Boltzmann test of §4 confirms this full density is *numerically necessary*—the acoustic peaks degrade by a factor of ~ 22 in RMS with the dynamical $\Omega_{m,\mathrm{dyn}} = 0.160$ alone. What remains open is a first-principles derivation of these algebraic forms from the field action; the burden has moved from "derive MIRA" to "ground ω_b and the ω_c identity." The values are not fitted, but their UV origin is open.
- **The φ-dark-matter mass (OP-9).** The mass $m_\varphi \simeq 41.02$ eV used in the extension sector follows from the algebraic relation $m_\varphi = \Sigma m_\nu^{\mathrm{active}}\,(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS})$, a dimensionally consistent product of a neutrino mass scale and a pure (φ, π) number. What remains open is the UV origin of the multiplier—a first-principles derivation from the field potential. The growth-sector predictions that descend from it—notably a free-streaming step in the matter power spectrum near $k_{\mathrm{fs}} \simeq 0.762\, h\,\mathrm{Mpc}^{-1}$, testable by DESI Y3 and Euclid—are deferred to future work and deliberately excluded from the core register of Table 3 until the multiplier is derived.
- **The $\beta_c = -\mathrm{AURA}$ coupling (OP-7).** The conformal coupling agrees with $-\mathrm{AURA} = -3.998$ to 0.2%. We present this as a near-coincidence to be tested, not as a proven identity, pending a QFT-level derivation of the role assignments.

The growth-of-structure status of §5 mirrors this structure: the single-sector core inherits an S_8 tension at the single-sector ceiling (3.5σ KiDS-1000, $S_8 = 0.846$), and the two-sector extension that resolves it ($S_8 = 0.759$, 0.01σ of KiDS-1000) rests on the m_φ multiplier whose UV origin is the open item above. The non-linear treatment required to settle the matter definitively demands N-body simulations beyond this work's scope. We record the core-level tension as an open challenge and the extension as falsifiable future-data territory (DESI Y3, Euclid).

None of these gaps touch the algebraic core of Table 3: the dark-energy equation of state, the spectral index, the tensor ratio, and the vanishing EFT slip functions stand on the dimensionless invariants alone and are falsifiable today.

8 Conclusions

We have presented a minimal-parameter framework in which the shape of the dark sector is fixed by the two dimensionless constants φ and π. The late-time equation of state emerges as bare ratios of structural invariants, $w_0 = -0.840$ and $w_a = -0.670$, and matches the DESI DR2 posterior at 0.16σ (Pantheon+) without a single fitted dark-sector parameter. The effective-field-theory description is canonical, with vanishing tensor, running, and braiding functions and $\alpha_K(0) = 0.4033$; the inflationary sector predicts $n_s = 0.9656$ and $r = 8.1 \times 10^{-3}$; and a Boltzmann calculation confirms that the full ω_m-direct matter density (the two-Ω structure) is physically required rather than cosmetic.

The framework's strength is its frugality and its consequent falsifiability. Its core predictions—the dark-energy ratios, the spectral index, the tensor ratio, and the vanishing slip functions—rest on φ and π alone and will be tested decisively by DESI Y3, Euclid, LiteBIRD, and CMB-S4 within this decade. We have been equally explicit about its limits: the vacuum scale is a measured input, the S_8 growth amplitude remains in tension with weak lensing, and the algebraic origin of the CMB matter density (ω_b, ω_c) together with the φ-dark-matter mass coefficient are unresolved. These define a concrete research agenda rather than fatal flaws, and none of them touch the algebraic core.

Whether the deep agreement between a handful of golden-ratio combinations and the measured dark sector reflects an underlying structural principle or a remarkable set of coincidences is precisely the question that the prediction register of §6 is designed to settle. The next generation of cosmological surveys will provide the answer.

A The two lineage laws of the constant dictionary

The look-elsewhere test of §2.3 rests on the dictionary being a *closed, finite* set fixed in advance, rather than an open search space. This appendix states the generative rules that make it so. We present them as a descriptive grammar of the Genesis 5.12 set [Almeida Vallejo, 2026]: they reproduce the dictionary exactly, but their *dynamical* origin is not claimed and remains the open mechanism of §7. Stating them explicitly is what turns "31 named invariants" from an assertion into a checkable construction.

Seeds and scaffold. Everything descends from two dimensionless seeds, $\varphi = (1+\sqrt{5})/2$ (UV) and π (IR). Their sum is the scaffold $\Omega = \varphi + \pi = 4.75962$, and their mean is the bifurcation point $\beta \equiv \mathrm{BIAL} = (\varphi + \pi)/2 = 2.37981$ at which the dictionary splits into two branches.

Law I — the copy law (φ-branch). The φ-branch is generated by a single root, the screening amplitude $\mathrm{AURA} = \beta + \varphi = 3.99784$, which propagates *only* by integer multiples and the dyadic half. These are the dimensional ceilings:

$$\mathrm{MIRA} = \tfrac{1}{2}\mathrm{AURA}, \quad \mathrm{AURA}, \quad \mathrm{DUAL} = 2\,\mathrm{AURA}, \quad \mathrm{TRIAL} = 3\,\mathrm{AURA} = T_r, \quad \mathrm{CUARTAL} = 4\,\mathrm{AURA}. \tag{13}$$

The same law acts on the scaffold itself, giving the denominators $K_v = 2\Omega$ and $M_v = 3\Omega$. The economy noted in §2.3 is a direct consequence: since $T_r = 3\,\mathrm{AURA}$ and $M_v = 3\Omega$, the common factor of three cancels in

$$|w_0| = \frac{T_r}{M_v} = \frac{3\,\mathrm{AURA}}{3\,\Omega} = \frac{\mathrm{AURA}}{\Omega} = 0.83996, \tag{14}$$

so the equation-of-state numerator and denominator are not two independent draws but a single ratio of the two branch roots, AURA/Ω.

Law II — the no-self-sum law (π-branch). The π-branch is generated additively: each child is a sum of previously defined parents, subject to the constraint that *a node never adds to itself.* Thus KAL $= \beta + \pi$, $P_{\rm sc}$ (PYROS) $= \Omega + \varphi$, IGNIS $= \pi + P_{\rm sc}$, and so on, building up to the convergences. A corollary is that two distinct lineages may land on the *same value* without being the same entity: by construction IGNIS $= K_v = 9.51925$ (disruptive vs. structural lineage) and SOLAR = MAR = 7.90122. The look-elsewhere count of §2.3 treats these honestly, deduplicating by value so the 31 named nodes collapse to 20 distinct numbers. The second equation-of-state ratio lives entirely on this branch,

$$|w_a| = \frac{P_{\rm sc}}{K_v} = \frac{\rm PYROS}{\rm IGNIS} = 0.67000, \tag{15}$$

where the denominator is read as the π-branch node IGNIS rather than the scaffold copy K_v—the two coincide in value by Law II.

Closure. The set is finite because both laws terminate at the five-dimensional integration ceiling $M_v = \varphi + \pi + K_v = 3\Omega = 14.27887$, the maximal dimensional invariant; no node exceeds it. This termination—not a choice made to fit data—is what bounds the dictionary to a fixed, enumerable set and licenses the conservative look-elsewhere denominator of §2.3. Any future extension would proceed only by the same copy law (the planned dimensional copies QUINTAL–DECAL), which, as noted in Table 1, add no accidental hit within the observational band of either w_0 or w_a.

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